

CHARA, Russian Federation

WMO: 303720

Lat: 56.900N

Lon: 118.267E

Elev: 711

StdP: 93.07

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.6	(c) -42.7	(d) -46.9	(e) 0.0	(f) -44.3	(g) -45.0	(h) 0.0	(i) -42.7	(j) 6.5	(k) -17.5	(l) 5.4	(m) -20.2	(n) 0.1	(o) 270	(p) 0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	27.9	16.8	26.0	16.2	24.0	15.6	18.5	24.5	17.5	23.3	16.5	22.2	2.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	12.8	20.3	15.4	11.9	19.3	14.4	11.1	18.6	55.5	24.4	52.1	23.3	49.2	22.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.1	5.1	-46.7	30.9	2.4	1.4	-48.5	31.9	-49.9	32.7	-51.3	33.5	-53.1	34.5
			-46.8	20.3	2.4	1.4	-48.5	21.3	-49.9	22.1	-51.3	22.9	-53.1	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.4	-31.2	-26.0	-16.5	-3.9	5.5	14.0	16.7	13.6	5.5	-5.4	-20.4	-30.3
	DBStd	17.97	6.09	6.49	7.28	5.22	3.88	3.92	3.07	3.44	4.02	5.89	7.93	6.29
	HDD10.0	6474	1277	1007	823	416	146	11	1	10	143	478	912	1249
	HDD18.3	9062	1536	1241	1081	666	397	134	68	148	385	737	1162	1507
	CDD10.0	478	0	0	0	0	7	132	208	123	8	0	0	0
	CDD18.3	24	0	0	0	0	0	6	16	3	0	0	0	0
	CDH23.3	538	0	0	0	0	3	168	273	93	2	0	0	0
	CDH26.7	104	0	0	0	0	0	33	59	12	0	0	0	0
(14) Wind	WSAvg	1.5	0.6	0.9	1.6	2.4	2.6	2.0	1.7	1.6	1.8	1.6	1.0	0.6
(15) Precipitation	PrecAvg	346	4	3	4	14	37	74	66	72	49	16	5	4
	PrecMax	481	14	9	22	51	88	138	138	171	102	81	22	13
	PrecMin	177	0	0	0	2	10	6	7	9	18	1	0	0
	PrecStd	74	3	3	4	11	19	38	33	37	20	15	4	3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.1	-4.4	5.2	13.5	23.0	29.9	30.5	28.3	22.0	11.0	3.4	-8.5
		MCWB	-12.5	-7.0	0.6	5.3	11.7	16.6	17.9	17.0	12.5	5.0	0.1	-9.9
	2%	DB	-15.2	-8.2	1.2	9.8	19.3	27.4	28.3	25.9	18.6	7.4	-1.2	-13.4
		MCWB	-16.2	-10.0	-2.5	3.3	9.7	15.8	17.4	16.5	11.2	2.9	-3.5	-14.4
	5%	DB	-18.8	-11.7	-1.4	7.2	16.5	25.3	26.4	23.7	15.7	4.9	-4.9	-17.5
		MCWB	-19.5	-13.0	-4.4	1.8	8.1	15.0	16.9	15.7	9.6	1.0	-6.5	-18.0
	10%	DB	-22.0	-14.7	-3.7	4.9	14.0	23.1	24.5	21.4	13.1	2.8	-8.4	-21.0
		MCWB	-22.4	-15.7	-6.2	0.2	6.8	14.1	16.5	14.7	8.3	-0.2	-9.5	-21.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.5	-6.8	0.4	5.6	12.1	18.7	19.9	19.2	14.0	5.2	0.2	-10.0
		MCDB	-11.3	-4.5	4.8	11.9	21.2	26.9	26.0	24.5	20.3	10.1	3.4	-8.7
	2%	WB	-16.1	-10.0	-2.4	3.6	10.4	16.9	18.9	17.6	12.0	3.2	-3.4	-14.4
		MCDB	-15.1	-8.3	0.9	9.0	17.9	24.4	24.9	23.1	16.3	6.9	-1.5	-13.5
	5%	WB	-19.4	-12.9	-4.4	2.0	8.7	15.9	18.1	16.5	10.4	1.5	-6.5	-18.0
		MCDB	-18.9	-11.7	-1.5	6.8	15.1	23.0	23.8	21.7	14.8	4.4	-4.9	-17.5
	10%	WB	-22.4	-15.7	-6.3	0.5	7.3	14.9	17.2	15.5	8.8	0.0	-9.5	-21.3
		MCDB	-22.0	-14.7	-3.8	4.3	13.1	21.4	22.9	20.3	12.7	2.5	-8.4	-21.0
(35) Mean Daily Temperature Range	MDBR		11.5	16.1	17.7	12.8	11.3	12.9	12.3	12.5	11.0	10.8	11.4	9.8
	5% DB	MCDBR	13.5	16.6	17.8	15.0	17.4	18.2	17.4	18.1	16.8	12.3	12.3	12.4
		MCWBR	12.7	15.0	14.3	9.1	8.8	7.8	7.7	9.2	9.4	8.0	10.1	11.7
	5% WB	MCDBR	13.5	16.5	17.4	14.0	15.5	15.3	13.7	14.4	14.9	11.5	12.1	12.3
		MCWBR	12.8	14.9	14.0	8.8	8.3	7.1	6.6	7.7	8.6	7.7	10.0	11.6
(40) Clear-Sky Solar Irradiance	taub		0.237	0.270	0.297	0.363	0.440	0.439	0.439	0.396	0.329	0.285	0.247	0.225
	taud		2.067	2.074	2.067	1.983	1.963	2.167	2.206	2.316	2.429	2.312	2.097	2.006
	Ebn at Noon		641	773	850	839	793	803	794	807	823	762	615	533
	Edh at Noon		57	89	117	151	166	138	130	109	83	70	55	45
(44) All-Sky Solar Radiation	RadAvg		0.77	1.73	3.32	5.02	5.39	5.24	4.87	4.03	2.70	1.68	0.94	0.53
	RadStd		0.04	0.08	0.20	0.34	0.52	0.39	0.44	0.31	0.25	0.12	0.08	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page